Computational Validation of Adaptive Interface Control in TMI Quantum Systems

Расширенный отчёт v5 по вычислительным тестам 1–28

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Аннотация

Этот отчёт обновляет вычислительную линию ТМИ до Tests 1–28. Версия v5 добавляет Test 28: out-of-distribution Qiskit-Aer generalization на новых схемах, которых не было в калибровке. Главный новый результат:

$$Calibrate(I_{sim}) \wedge Correct(I_{circuit}) \Rightarrow Generalize(C_{new}).$$

Это симуляторная проверка, не лабораторное доказательство физической теории.

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1. Статус отчёта

TMI Quantum Computational Validation v0.5

2. Summary of Tests 1–28

Test	Name	Criterion / meaning	Result
1	Ramsey toy model	T2_TMI > T2_standard	Delta T2 = 58.1 us
2	Noisy Ramsey learning	T2_TMI > T2_standard	Delta T2 = 39.3 us
3	Small quantum circuit	P_TMI > P_standard	Delta P = 0.115
4	Stabilized interface	Stable adaptive control	Delta P = 0.026
5	Surrogate Monte Carlo	E[P_TMI] > E[P_standard]	Delta = 0.0510, Pr = 0.864
6	Ablation study	Structured adaptive control	full TMI Delta = 0.0680
7	Density-matrix Monte Carlo	Full matrix validation	Delta = 0.0930, Pr = 0.933
8	Cross-model validation	Multiple circuit families	Delta = 0.0660, Pr = 0.846
9	Failure-boundary map	Weak/failure regions	failure = 0, weak = 6
10	Adversarial failure	Forced failure modes	failure cells = 58
11	Safety recovery	Mitigation after failure	recovered = 31/58
12	Reproducibility harness	One-command validation	passed = 11/11
13	Independent simulator	Independent simulator validation	Delta = 0.1121, Pr = 1.000
14	Noise-model stress	Noise class robustness	passed = 3/4, Delta = 0.0068
15	Amplitude damping recovery	Relaxation-aware layer	Delta_relax = 0.1532
16	Noise classifier	Classify then control	accuracy = 0.952, Delta = 0.0554
17	Online noise drift	Dynamic reclassification	Delta = 0.0255, acc = 0.816
18	Delayed feedback	Latency hurts online control	online = 0.883, delayed d2 = 0.861
19	Latency policy optimization	Predictive latency layer	opt = 0.875 > d2 = 0.861

Test	Name	Criterion / meaning	Result
20	Cost-overhead tradeoff	Utility and efficiency	U_highperf = 0.0705
21	Cost-interface boundary	B_cost	lambda* = 0.0636
22	Multi-objective control	Boundary manifold	standard share = 0.608, cost-aware-low = 0.373
23	Standard simulator protocol	Backend-independent Qiskit-style protocol	Delta = 0.0414, Pr = 0.750
24	Raw Qiskit-Aer execution	External simulator without calibration	failed, Delta = -0.00097, cells = 0/16
25	Qiskit noise calibration	Simulator-interface calibration	54 candidate calibrations; Loss_sim = 0.00106
26	Calibrated Qiskit-Aer validation	Held-out calibrated validation	passed, Delta = 0.0725, cells = 12/16
27	Circuit-inverse audit	Correct inverse and rerun Qiskit-Aer	passed, Delta = 0.1144, cells = 16/16
28	OOD Qiskit-Aer generalization	Unseen circuit families	passed, Delta = 0.1252 to 0.1414, cells = 24/24

3. External Qiskit-Aer branch: failure, calibration, correction, generalization

The external branch now has the following structure:

$$Test\,24:\ failed \rightarrow Test\,25:\ calibrated \rightarrow Test\,26:\ passed \rightarrow Test\,27:\ corrected \rightarrow Test\,28:\ OOD\ pa$$

3.1. Test 24: raw Qiskit-Aer failure

$$P_{base} = 0.7535, \quad P_{standard} = 0.7532, \quad P_{TMI} = 0.7523.$$
$$\Delta(P_{TMI} - P_{standard}) = -0.00097, \quad cells = 0/16.$$

3.2. Test 25: simulator-interface calibration

Test 25 found 54 candidate calibrations and introduced the diagnostic concept:

$$I_{sim} : TMI\ model \rightarrow Qiskit\ NoiseModel, \qquad Loss_{sim}(I_{sim}) > 0.$$

In the default-like calibrated regime:

$$\Delta(P_{standard} - P_{base}) = 0.0191, \quad \Delta(P_{TMI} - P_{standard}) = 0.0299.$$

3.3. Test 26: calibrated Qiskit-Aer validation

$$P_{base} = 0.4793, \quad P_{standard} = 0.5203, \quad P_{TMI} = 0.5928.$$
$$\Delta(P_{TMI} - P_{standard}) = 0.0725, \quad cells = 12/16.$$

3.4. Test 27: circuit-inverse audit

Original circuits failed ideal inverse correctness; corrected circuits passed. The key fix was:

$$\texttt{S} \to \texttt{Sdg}, \qquad S^{-1} = S^{\dagger}.$$

After correction:

$$P_{base} = 0.5791, \quad P_{standard} = 0.6432, \quad P_{TMI} = 0.7576.$$

$$\Delta(P_{TMI} - P_{standard}) = 0.1144, \quad cells = 16/16.$$

3.5. Test 28: out-of-distribution Qiskit-Aer generalization

Test 28 used unseen circuit families, not used in calibration or correction. The conservative uploaded profile gives:

$$P_{base} = 0.2914, \quad P_{standard} = 0.3424, \quad P_{TMI} = 0.4616.$$

$$\Delta(P_{standard} - P_{base}) = 0.0511, \quad \Delta(P_{TMI} - P_{standard}) = 0.1191.$$
$$Pr(P_{TMI} > P_{standard}) = 1.0, \quad cells = 24/24.$$

Across reported OOD profiles, all passed:

$$\texttt{best\_by\_channel}, \quad \texttt{conservative}, \quad \texttt{stress\_high\_signal}.$$

4. Visual summary

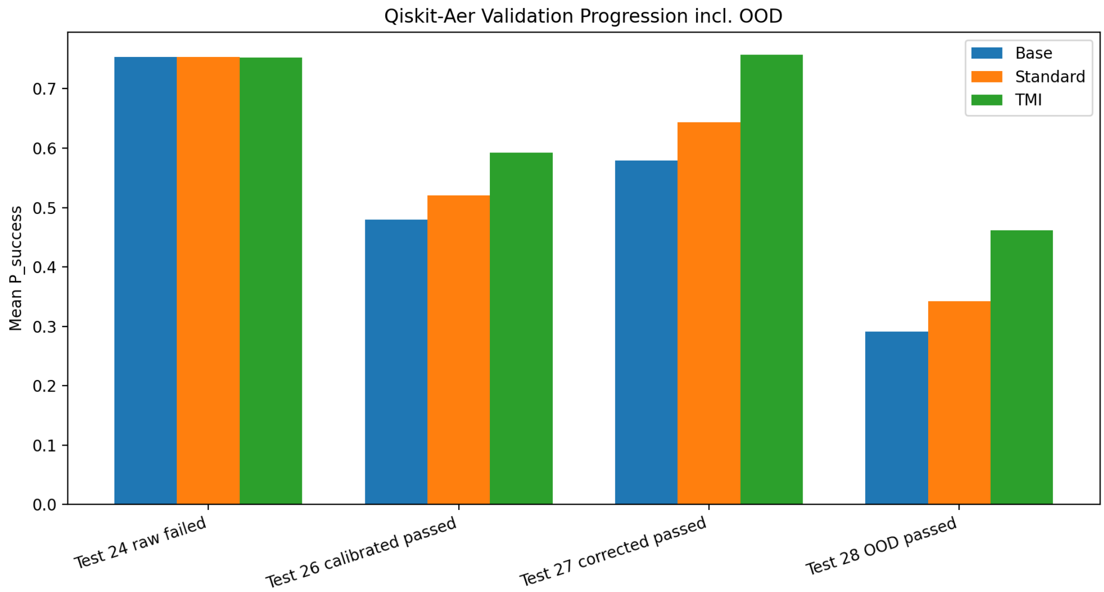

Рис. 1: External Qiskit-Aer progression including OOD generalization.

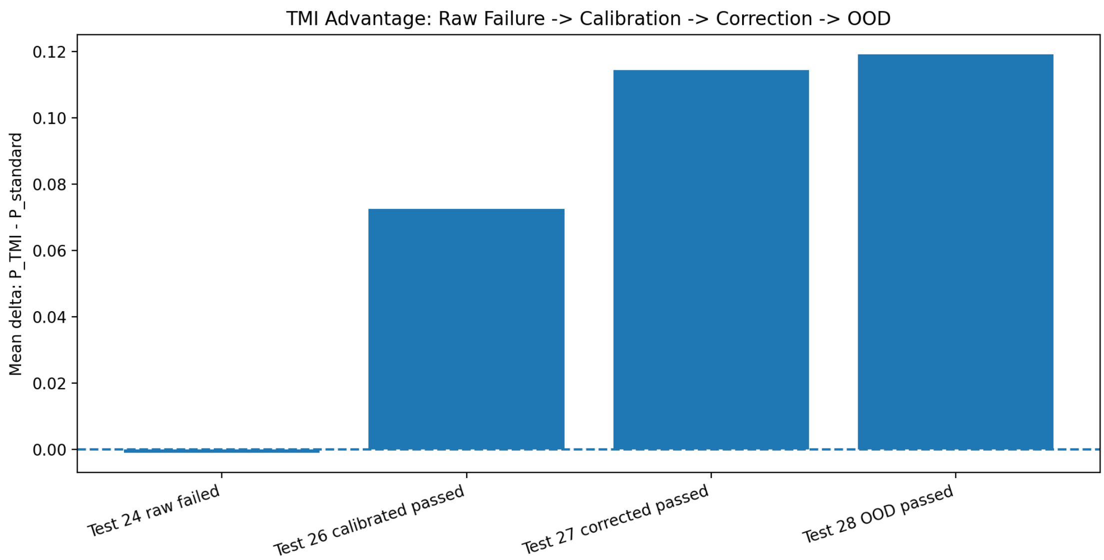

Рис. 2: $P_{TMI} - P_{standard}$: raw failure becomes positive after calibration, correction and OOD validation.

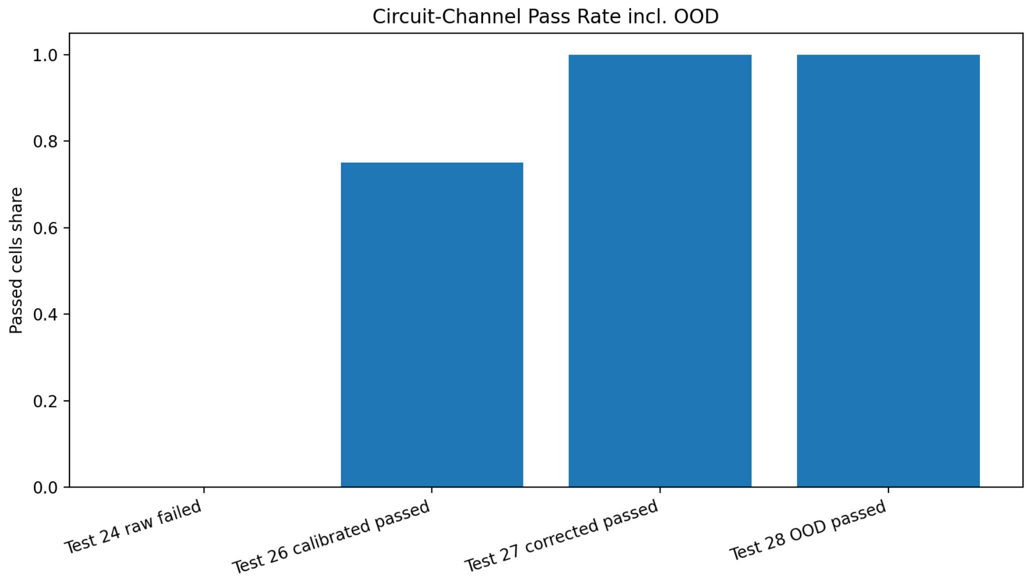

Рис. 3: Cell pass rate: 0/16, 12/16, 16/16, 24/24.

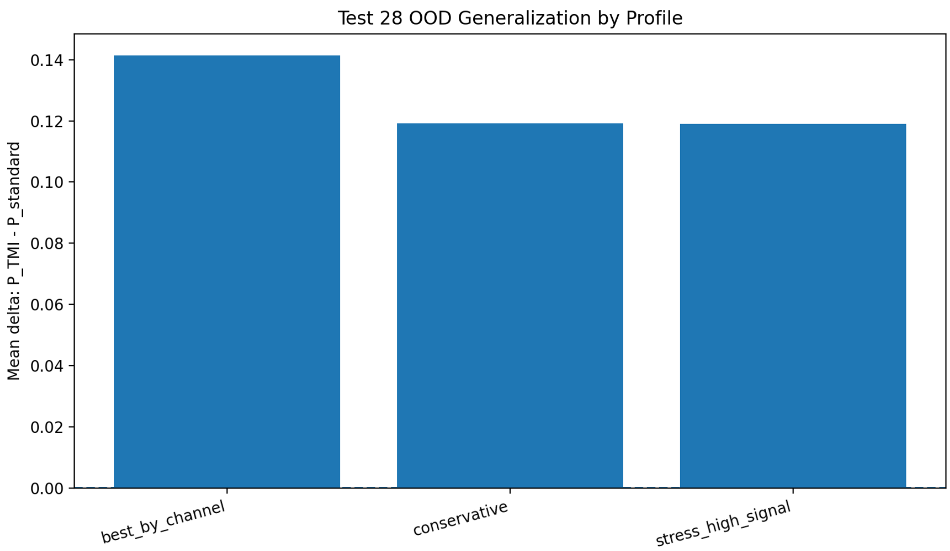

Рис. 4: Test 28 OOD deltas by profile.

5. v5 formulas

$$\boxed{Calibrate(I_{sim}) \wedge Correct(I_{circuit}) \Rightarrow Generalize(C_{new})}$$

$$\boxed{C_{new} \notin C_{calibration} \Rightarrow P_{TMI} > P_{standard} > P_{base}}$$

6. Conclusion

Version v5 adds out-of-distribution Qiskit-Aer generalization. The result is stronger than v4 because it shows that the calibrated/corrected simulator-interface mechanism is not limited to the circuit families used in calibration.

A. Состав пакета

Пакет содержит PDF и LaTeX исходники на русском и английском, JSON/CSV summary, графики внешней Qiskit-Aer ветки, пакеты Tests 12–28 и статусные файлы Tests 24–28.